Quick scheme + 1st order Upwind scheme to solve 2D Temeperature problem

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Problem statement

Given a 2 dimensional square domain of Length L = $1m$.
The Governing equation for Temperature reads :

$$\frac{\partial}{\partial x}(\rho u T) + \frac{\partial}{\partial y}(\rho v T) = \frac{\partial}{\partial x}(k\frac{\partial T}{\partial x}) + \frac{\partial}{\partial y}(k\frac{\partial T}{\partial y}) \tag{1}$$

where : $\rho = cst$, u, v are uniform on the domain and known , k is the thermal conductivity , T is the temperature.
Boundary conditions are given to be dirichlet boundary conditions :

- $T_{left} = 4$
- $T_{Top} = 1$
- $T_{right} = 0$
- $T_{bottom} = 0$

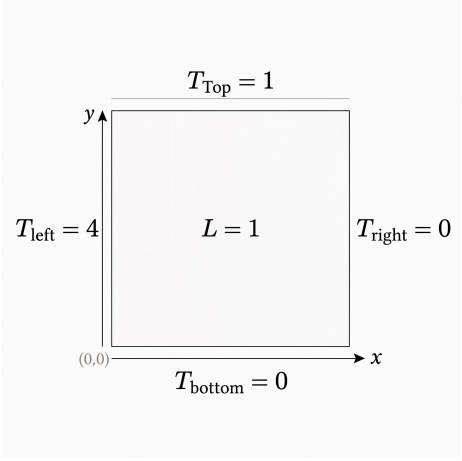


Figure 1: 2D square domain

Discretization scheme

We want to have a discretization with uniform mesh i.e $h_x = h_y = h$
Vertex centered FVM is used to construct 2D grid of $N = 21$ x 21 Nodes/Control volumes. Grid spacing $h = \frac{L}{N-1}$. Assuming that $A_e = A_w = A_n = A_s = A$. ie a uniform crossectional area of the meshes.

Integrate the governing equation (1) onto a finite volume centered at P and using Gauss theorem we obtain the flux equation:

$$F_e T_e - F_w T_w + F_n T_n - F_s T_s = k_e \frac{\partial}{\partial x} T_e - k_w \frac{\partial}{\partial x} T_w + k_n \frac{\partial}{\partial y} T_n - k_s \frac{\partial}{\partial y} T_s$$

(Flux equation)

Where:

- F = Convection strength =convective transport coefficient = $\rho u, \rho v$
- D = Deffusive strength =diffusive transport coefficient = k/h

$F_e = F_w = F_x = \rho u$, $F_n = F_s = F_y = \rho v$ due to conservativeness .

$k_e = k_w = k_n = k_s$

Assuming that T varies piecewise Linear between nodes we have :

1. $\frac{\partial T}{\partial x}\big|_e = \frac{T_E - T_P}{h}$
2. $\frac{\partial T}{\partial x}\big|_w = \frac{T_P - T_W}{h}$
3. $\frac{\partial T}{\partial y}\Big|_n = \frac{T_N - T_P}{h}$
4. $\frac{\partial T}{\partial y}\Big|_s = \frac{T_P - T_S}{h}$

Since we have uniform meshes and k is uniform it implies that $D_e = D_w = D_n = D_s = D$
Therefore:

$$F_x(T_e - T_w) + F_y(T_n - T_s) = k[(\frac{T_E - T_P}{h}) - (\frac{T_P - T_W}{h}) + (\frac{T_N - T_P}{h}) - (\frac{T_P - T_S}{h})]$$

$$F_x(T_e - T_w) + F_y(T_n - T_s) = D[(T_E - T_P) - (T_P - T_W) + (T_N - T_P) - (T_P - T_S)]$$

$$Pe_x(T_e - T_w) + Pe_y(T_n - T_s) = (T_E - T_P) - (T_P - T_W) + (T_N - T_P) - (T_P - T_S) \quad (2)$$

where Pe_x and Pe_y are the grid peclet numbers in x and y respectively.

$Pe_x = \frac{F_x}{D}$ and $Pe_y = \frac{F_y}{D}$

Consider $u > 0, v > 0$ PLUS Consider unifrom grid , same distance between grid nodes in x and in y and using Vertex centered FVM to generate 2D grid of points.

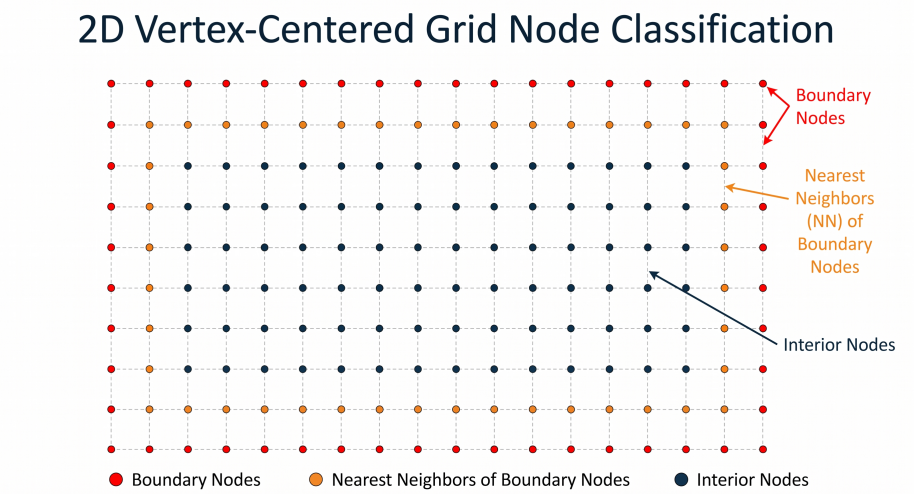


Figure 2: 2D GRID

We will use Quick scheme for interior Nodes i.e grid points that are not NN of boundary nodes.
east face Stencil : W P | E

$$T_e = \frac{3}{8}T_E + \frac{6}{8}T_P - \frac{1}{8}T_W \tag{3}$$

north face : Calculation of east face maps to north face because of same weights due to uniform grid, just change Stencil : S, P | N

$$T_n = \frac{3}{8}T_N + \frac{6}{8}T_P - \frac{1}{8}T_S \tag{4}$$

west face Stencil : WW W | P

$$T_w = \frac{3}{8}T_P + \frac{6}{8}T_W - \frac{1}{8}T_{WW} \tag{5}$$

south face : calculation of west face maps to south face because of same weights due to uniform grid, just change stencil : SS S | P

$$T_s = \frac{3}{8}T_P + \frac{6}{8}T_S - \frac{1}{8}T_{SS} \tag{6}$$

EQN (2) after Substituting the face values using Quick scheme

$$\begin{aligned} Pe_x(\frac{3}{8}T_E + \frac{3}{4}T_P - \frac{1}{8}T_W - [\frac{3}{8}T_P + \frac{3}{4}T_W - \frac{1}{8}T_{WW}]) \\ +Pe_y(\frac{3}{8}T_N + \frac{3}{4}T_P - \frac{1}{8}T_S - [\frac{3}{8}T_P + \frac{3}{4}T_S - \frac{1}{8}T_{SS}]) = \\ (T_E - T_P) - (T_P - T_W) + (T_N - T_P) - (T_P - T_S) \end{aligned}$$

Resulting in :

$$\begin{aligned} Pe_x(\frac{3}{8}T_E + \frac{3}{8}T_P - \frac{7}{8}T_W + \frac{1}{8}T_{WW}) \\ +Pe_y(\frac{3}{8}T_N + \frac{3}{8}T_P - \frac{7}{8}T_S + \frac{1}{8}T_{SS}) = \\ T_E - 4T_P + T_W + T_N + T_S \end{aligned}$$

$$\begin{aligned} (4 + \frac{3}{8}Pe_x + \frac{3}{8}Pe_y)T_P = \\ (1 - \frac{3}{8}Pe_x)T_E + (1 + \frac{7}{8}Pe_x)T_W + \\ (1 - \frac{3}{8}Pe_y)T_N - \frac{Pe_x}{8}T_{WW} + 1 + \frac{7}{8}Pe_y)T_S - \frac{Pe_y}{8}T_{SS} \end{aligned}$$

$$a_P T_P = a_E T_E + a_W T_W + a_N T_N + a_{WW} T_{WW} + a_S T_S + a_{SS} T_{SS} \qquad (7)$$

with :

1. $a_E = 1 - \frac{3}{8}Pe_x$
2. $a_W = 1 + \frac{7}{8}Pe_x$
3. $a_N = 1 - \frac{3}{8}Pe_y$
4. $a_S = 1 + \frac{7}{8}Pe_y$
5. $a_{WW} = -\frac{Pe_x}{8}$
6. $a_{SS} = -\frac{Pe_y}{8}$
7. $a_P = a_E + a_W + a_N + a_S + a_{WW} + a_{SS}$

For boundedness of the scheme we require that :

$0 \leq \frac{a_W}{a_P} \leq 1$, $0 \leq \frac{a_E}{a_P} \leq 1$, $0 \leq \frac{a_N}{a_P} \leq 1$, $0 \leq \frac{a_S}{a_P} \leq 1$,$0 \leq \frac{a_{WW}}{a_P} \leq 1$,$0 \leq \frac{a_{SS}}{a_P} \leq 1$

We will use 1st order Upwind scheme for NN of boundary points i.e grid points that are nearest to the boundary nodes.

east face:

$$T_e = T_P \qquad (8)$$

north face:

$$T_n = T_P \tag{9}$$

west face:

$$T_w = T_W \tag{10}$$

south face :

$$T_s = T_S \tag{11}$$

EQN (2) after Substituting the face values using 1st order Upwind scheme

$$Pe_x(T_P - T_W) + Pe_y(T_P - T_S) = (T_E - T_P) - (T_P - T_W) + (T_N - T_P) - (T_P - T_S)$$

Resulting in :

$$Pe_x(T_P - T_W) + Pe_y(T_P - T_S) = T_E - 4T_P + T_W + T_N + T_S$$

Therefore:

$$(4 + Pe_x + Pe_y)T_P = T_E + (1 + Pe_x)T_W + T_N + (1 + Pe_y)T_S$$

We can rewrite the eqn to :

$$a'_P T_P = a'_E T_E + a'_W T_W + a'_N T_N + a'_S T_S \tag{12}$$

with :

1. $a'_E = 1$
2. $a'_W = 1 + Pe_x$
3. $a'_N = 1$
4. $a'_S = 1 + Pe_y$
5. $a'_P = a'_E + a'_W + a'_N + a'_S$

For boundedness of the scheme we require that :

$0 \leq \frac{a'_W}{a'_P} \leq 1$, $0 \leq \frac{a'_E}{a'_P} \leq 1$, $0 \leq \frac{a'_N}{a'_P} \leq 1$, $0 \leq \frac{a'_S}{a'_P} \leq 1$

Numerical Results

1. Speed of 300 ms^{-1} : Pe in x = 3.75 , Pe in y= 3.75 Temperature distribution obtained after 176 iterations

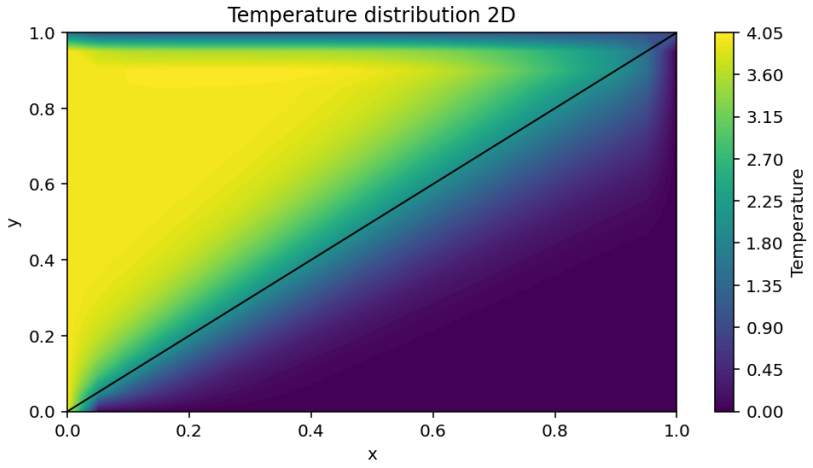


Figure 3: 2D Temperature distribution for high velocity profile 300 ms^{-1}

2. Speed of 150 ms^{-1} : Pe in x = 1.875 , Pe in y= 1.875 Temperature distribution obtained after 120 iterations

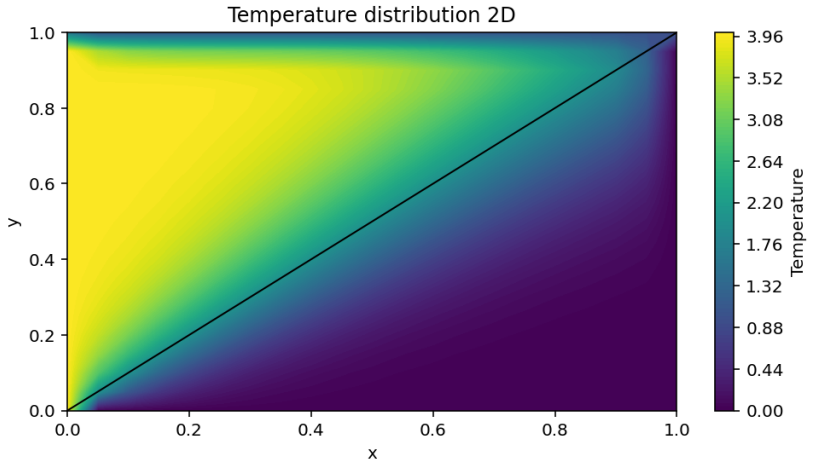


Figure 4: 2D Temperature distribution for high velocity profile 150 ms^{-1}

3. Speed of 50 ms^{-1} : Pe in x = 0.625 , Pe in y= 0.625 Temperature distribution obtained after 1027 iterations

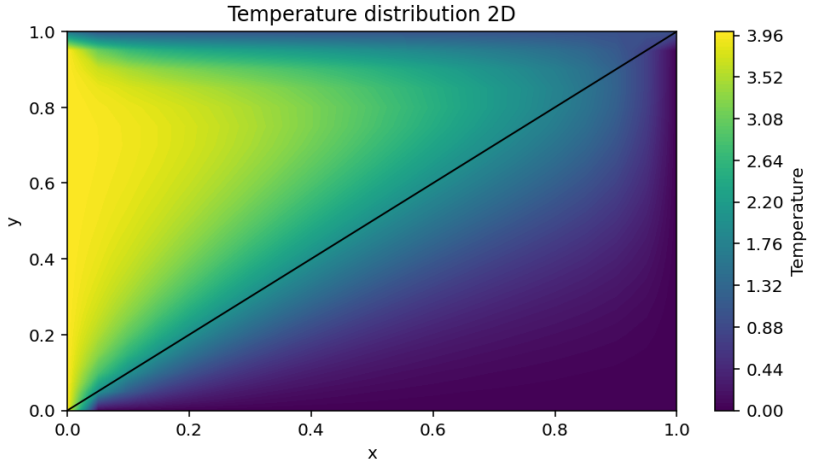


Figure 5: 2D Temperature distribution for high velocity profile 50 ms^{-1}

4. Speed of 10 ms^{-1} : Pe in x = 0.125 , Pe in y= 0.125 Temperature distribution obtained after 9868 iterations

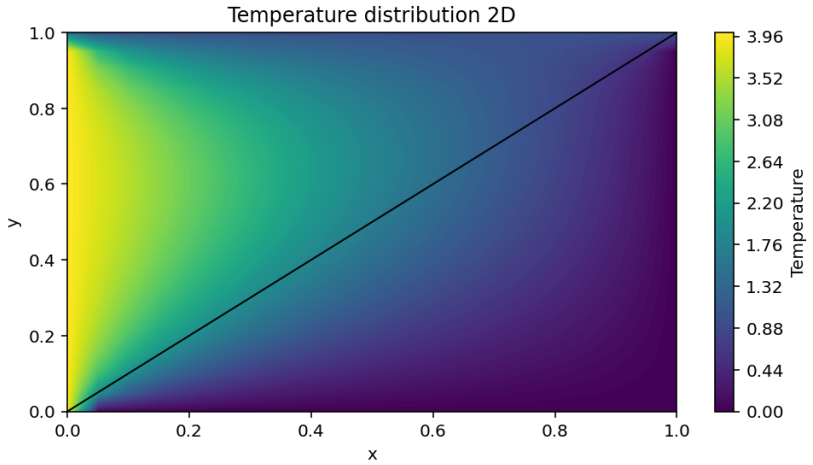


Figure 6: 2D Temperature distribution for high velocity profile 10 ms^{-1}

5. Speed of 5 ms^{-1} : Pe in x = 0.125 , Pe in y= 0.125 Temperature distribution obtained after 9868 iterations

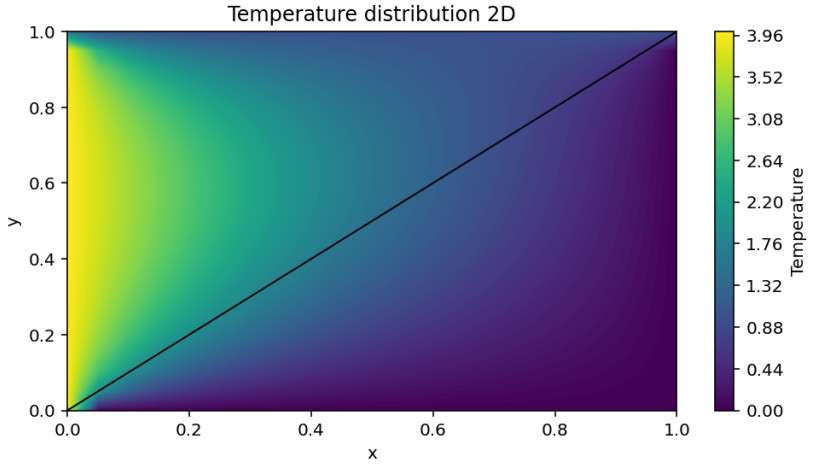


Figure 7: 2D Temperature distribution for high velocity profile 5 ms^{-1}

Discussion of Results

- QUICK scheme

  Investigating the boundedness condition for speed starting at 5 ms^{-1} up to 300 ms^{-1} for the QUICK scheme we can clearly see that some coefficients are negative and as the speed increases more and more coefficients become negative , 2 negative coeff. for speed of 5 vs 4 negative coeff. for speed of 300.

	Case 1	Case 2	Case 3	Case 4	Case 5
Péclet number	3.75	1.875	0.625	0.125	0.0625
speed	300.0	150.0	50.0	10.0	5.0
aE	-0.059633	0.054913	0.171329	0.232824	0.241313
aW	0.62844	0.488439	0.346154	0.270992	0.260618
aN	-0.059633	0.054913	0.171329	0.232824	0.241313
aS	0.62844	0.488439	0.346154	0.270992	0.260618
aWW	-0.068807	-0.043353	-0.017483	-0.003817	-0.001931
aSS	-0.068807	-0.043353	-0.017483	-0.003817	-0.001931
aP	6.8125	5.40625	4.46875	4.09375	4.046875

Table 1: Quick Scheme Coefficients Validation check for boundedness

- 1st order Upwind Scheme

  Investigating the boundedness condition for speed starting at 5 ms^{-1} up to 300 ms^{-1}.The first order upwind scheme does not produce negative coefficients meaning that the solution produced using this scheme should be bounded by the upper and lower boundaries values.

	Case 1	Case 2	Case 3	Case 4	Case 5
Péclet number	3.75	1.875	0.625	0.125	0.0625
speed	300.0	150.0	50.0	10.0	5.0
aE	0.086957	0.129032	0.190476	0.235294	0.242424
aW	0.413043	0.370968	0.309524	0.264706	0.257576
aN	0.086957	0.129032	0.190476	0.235294	0.242424
aS	0.413043	0.370968	0.309524	0.264706	0.257576
aP	11.5	7.75	5.25	4.25	4.125

Table 2: Upwind Scheme Coefficients Validation check for boundedness

- Upon testing for speeds > 340 ms^{-1} the combined discretisation scheme (Quick and UDS1) fails to produce converging results and an Unbounded solution is obtained.

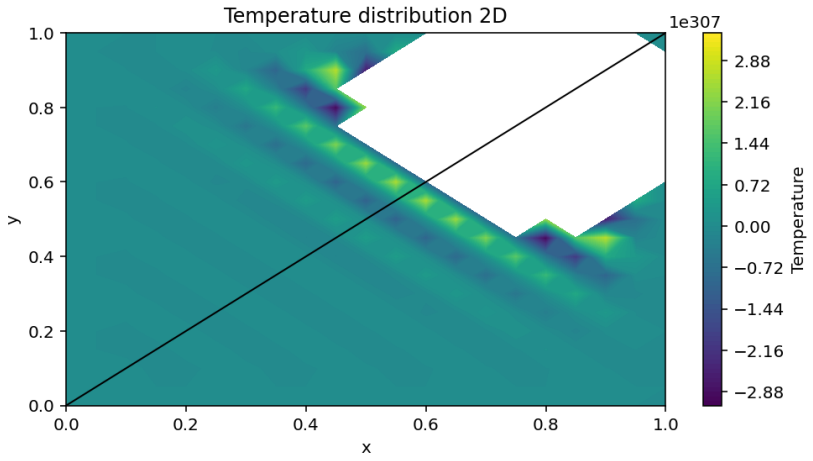


Figure 8: 2D Temperature distribution for high velocity profile 350 ms^{-1}

Conclusion

The combination of Quick scheme to treat interior nodes and first order Upwind to treat Nearest boundary nodes is accurate for velocities that are less than 330 ms^{-1} and it is possible to solve the convection and diffusion of temperature on a 2 dimensional domain. Above this velocity, the scheme no longer guarantees a bounded solution.